

MTH 377/577 Convex Optimization

Practice Quiz

Instructions: Answer all questions. Time: 60 minutes. Maximum Marks: 20

1. State whether the following statements are true/false. For true statements, provide a short proof. For False, provide a counter example.
 - (a) The set $A = \{(x_1, x_2, \dots, x_n) \in R^n, |\sum_i x_i \leq n\}$ is compact. (3)
 - (b) Consider a set $A = \{\theta_1 x_1 + \theta_2 x_2 | x_1 = (4, 5), x_2 = (1, 3); \forall \theta_1, \theta_2 \geq 0\}$. A is a convex cone. (2)
 - (c) The function $f(x) = -(x - 2)^2$ is convex. (2)
 - (d) The set $C = A \cup B$ where A, B are convex is always convex. (2)
2. Let $\{x_k\}_{k \in I_+}$ be a sequence in R where $x_{k+1} = (-1)^k x_k$. Let X be the set of all terms x_k in the given sequence. Let $Y = \{x \in R | x > 1\}$.
 - (a) Does there exist a hyperplane that separates X and Y ? If yes, find such a hyperplane. If not, why? Use separating hyperplane theorem(s) in support of your answer. (2.5)
 - (b) Let $f : X \rightarrow R$ and $g : Y \rightarrow R$ be continuous functions. Do f, g have a maxima/minima? Provide an explanation for your answer. (2.5)
3. Find the (i) convex hull, (ii) affine hull for the following set: (3)

$$A = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$$
4. Use Farkas' lemma and find out if the system $Ax = b$ has a non-negative solution when

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, x \in R^3$$

Show all the steps. (3)